

PRACTICAL - 6

Implementation of Matrix Chain Multiplication using Dynamic Programming

Aim of the Practical:

To find the parenthesization with minimum scalar multiplications using dynamic programming and derive its complexity.

Recurrence:

$$m[i, j] = 0, \text{ if } i = j$$

$$m[i, j] = \min\{i \leq k < j\} [m[i, k] + m[k+1, j] + p[i-1]p[k]p[j]]$$

Program:

```
#include <iostream>
#include <vector>
#include <climits>
using namespace std;
long long matrixChain(const vector<int>& p) {
    int n = p.size() - 1;
    vector<vector<long long>>> dp(n + 1, vector<long long>(n + 1, 0));
    for (int len = 2; len <= n; len++) {
        for (int i = 1; i <= n - len + 1; i++) {
            int j = i + len - 1;
            dp[i][j] = LLONG_MAX;
            for (int k = i; k < j; k++) {
                long long cost = dp[i][k] + dp[k + 1][j]
                    + 1LL * p[i - 1] * p[k] * p[j];
                if (cost < dp[i][j]) dp[i][j] = cost;
            }
        }
    }
    return dp[1][n];
}
int main() {
    vector<int> p = {4, 10, 3, 8};
    cout << matrixChain(p);
}
```

Output:

216

Step-by-Step Cost Calculation for the Sample:

- $A_1 = 4 \times 10$, $A_2 = 10 \times 3$, $A_3 = 3 \times 8$.
- $(A_1 A_2) A_3$: cost = $(4 \times 10 \times 3) + (4 \times 3 \times 8) = 120 + 96 = 216$.
- $A_1 (A_2 A_3)$: cost = $(10 \times 3 \times 8) + (4 \times 10 \times 8) = 240 + 320 = 560$.
- Minimum = 216 scalar multiplications.

Time Analysis:

Derivation of Time Complexity:

Let n be the number of matrices.

The outer loop chooses chain length len from 2 to n : $O(n)$ possibilities.

For a fixed len , the starting index i can take at most $O(n)$ values.

For every pair (i, j) , the split point k runs from i to $j-1$: at most $O(n)$ choices.

$$T(n) = \sum_{len} \sum_i \sum_k O(1)$$

$$T(n) = O(n \cdot n \cdot n) = O(n^3)$$

More precisely, the number of evaluated split points is:

$$\sum_{len=2 \text{ to } n} (n - len + 1)(len - 1)$$

which grows on the order of n^3 .

The DP table has $(n + 1)^2$ entries.

$$S(n) = \Theta(n^2)$$

Final Time Analysis:

- Time Complexity: $\Theta(n^3)$.
- Space Complexity: $\Theta(n^2)$.

Result:

Matrix Chain Multiplication was solved successfully. The sample minimum cost is 216 and the DP complexity is cubic in the number of matrices.